Print Management System

College of Computing and Mathematics

SRS Document

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Project Motivation

The primary motivation for this project is to replace the existing PaperCut printing solution used within the College of Computing and Mathematics. Although PaperCut meets basic printing needs, its recurring licensing and usage costs have become a financial burden, and its proprietary nature limits flexibility, transparency, and long-term institutional control.

The proposed Printing Management System aims to provide a cost-effective, internally owned alternative that eliminates recurring fees while giving the college full ownership of system data, features, and future enhancements. A key driver of this project is the need for improved visibility into printing usage and expenses through analytics and cost calculation capabilities, enabling informed decision-making and more effective cost management.

The system will serve all faculty members, staff members, and undergraduate and graduate students across all departments in the college. It will support multiple user roles, including administrators, technical support staff, and normal users, ensuring secure access, efficient management, and ease of use while reducing costs without compromising printing services.

Project Users

Administrator (Admin)

The Administrator is the highest-privileged user role in the Printing Management System and has full access to all system functionalities. The Admin is responsible for **adding and removing printing devices**, **deleting and configuring print queues**, **modifying user printing quotas**, **managing user access by suspending or restricting users in cases of misuse**, **accessing detailed user and device printing history logs**, **viewing and analyzing printing cost calculations**, and **accessing the analytics dashboard to review usage trends and system statistics**, ensuring effective control, accountability, and informed decision-making.

Technician

The Technician role has limited administrative access within the Printing Management System and is responsible for supporting day-to-day system operation. Technical Support staff can **manage (increase/decrease) user printing quotas**, **manage (suspend/unsuspend) user access**, and **receive system notifications related to errors and operational issues**, such as printer faults, low ink or paper conditions, and system failures. This role does not have permission to add or remove printing devices, ensuring separation of responsibilities while enabling timely issue resolution and user support

Standard User

The Standard User represents faculty members and students who use the Printing Management System for regular printing activities. This role allows users to **submit print jobs** and **view their own printing history logs**, providing visibility into personal printing activity while ensuring that users can only access their own data. Standard Users do not have administrative or system management privileges.

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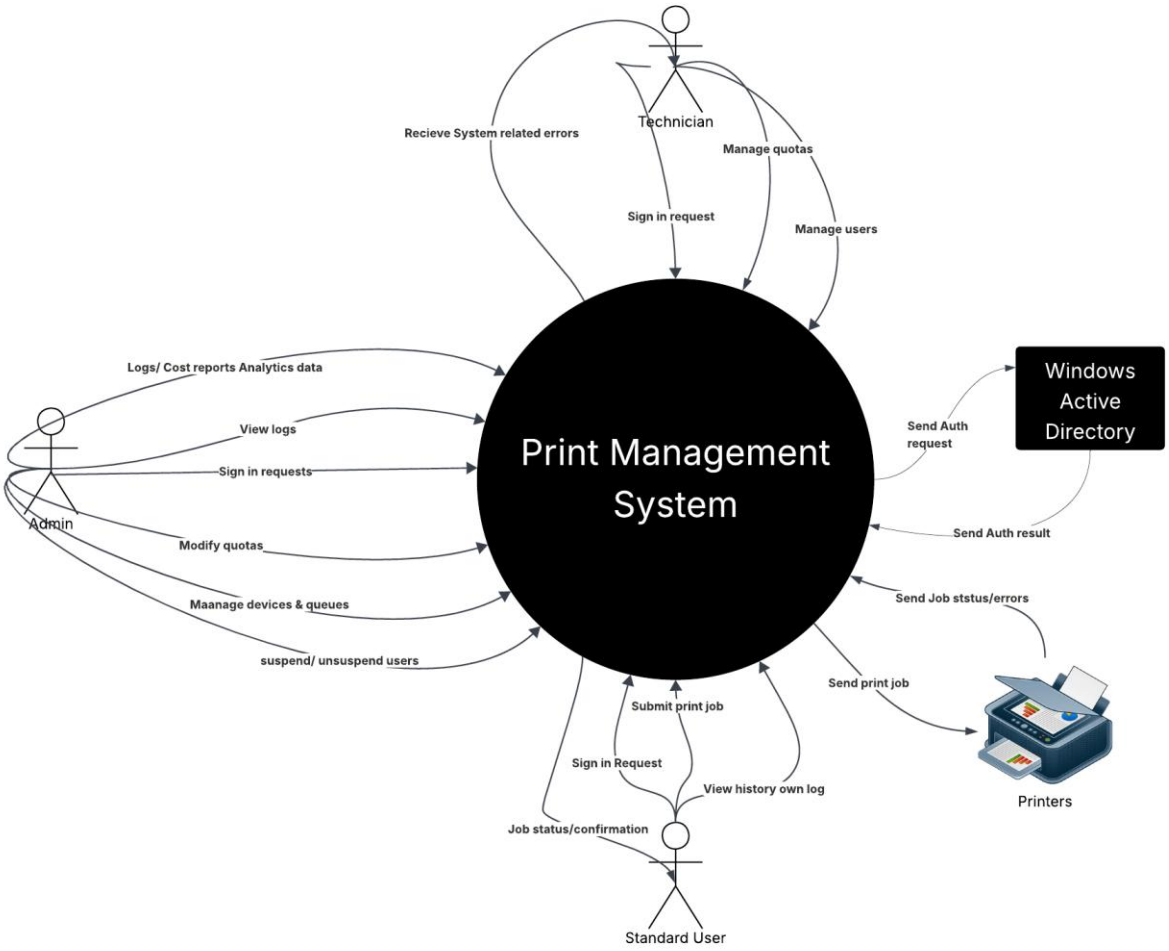
System Overview

System Context Diagram



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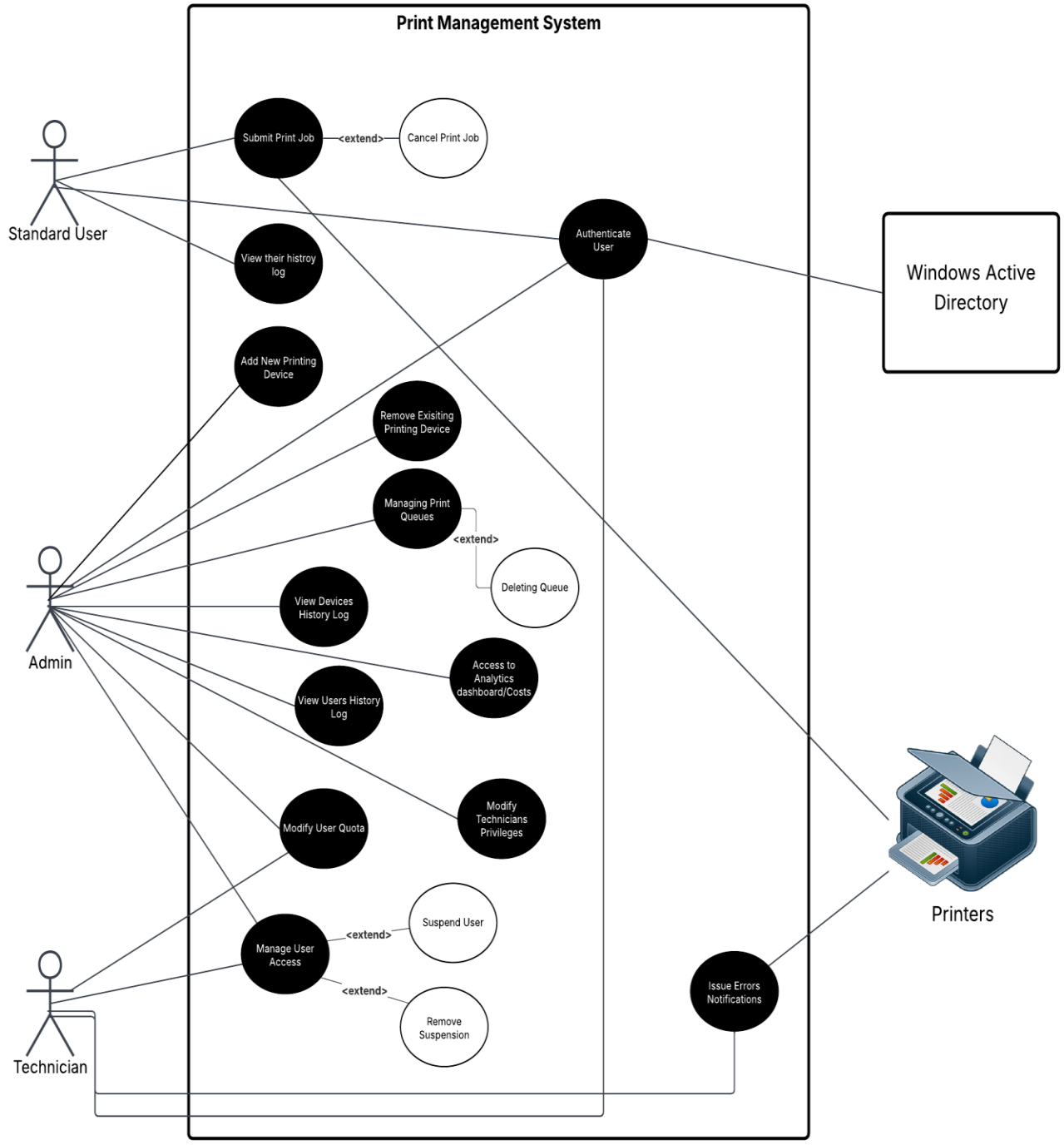
Use Case Diagram



Use Case Description

UC-01: Authenticate User

Id	UC-01
Title	Authenticate User
Actors	Standard User (Primary) / Administrator (Primary)/ Technical Support (Primary)/ Microsoft Active Directory (Secondary)
Goal	Verify the identity of a user through Windows Active Directory in order to grant authorized access to the system.
Preconditions	- The user has a valid Active Directory account. - The system is connected to the Active Directory service.
Postconditions	- The user is authenticated. - The system retrieves user identity attributes (e.g., username, department, group membership). - The system assigns appropriate role-based access permissions.
Basic Flow	1. User attempts to access the Printing Management System. 2. System prompts the user for credentials (or redirects to AD authentication service). 3. User provides AD credentials. 4. System sends authentication request to Active Directory. 5. Active Directory validates credentials. 6. Active Directory returns authentication result and user identity attributes. 7. System grants access based on the user's assigned role.
Alternate Flows	**A1 — Invalid Credentials:** If the credentials are invalid, AD rejects the request and the system denies

	access, displaying an error message. **A2 — AD Service Unavailable:** If the system cannot communicate with AD, authentication fails and the system displays a service error message. **A3 — Suspended User:** If the user account is marked as suspended within the system, access is denied even if AD authentication succeeds.
Notes	All authentication attempts shall be logged for security and auditing purposes. Role authorization shall be determined after successful authentication. This use case is included in all secured system use cases

UC-02: Request Print Job (Submit to Print Queue)

Id	UC-02
Title	Submit Print Job (Submit to Print Queue)
Primary Actor	Standard User
Goal	Submit a document to the printing system and have it accepted for printing.
Preconditions	-User is authenticated via Windows Active Directory and has an active account. -The user's account is not suspended. -At least one printing device/queue is available. -The User's quota is not exceeded
Postconditions	- A print job record is created in the system - The job is placed in the selected print queue and is either printed or queued for printing - User receives confirmation (or failure) and a job identifier. - The file must be deleted from the printing queue.
Basic Flow	8. User selects a document and chooses "Print" in the client or web

	interface. 9. User selects target printer or queue and print options (copies, color/BW, pages). 10. System validates user's printing quota and print options. 11. If quota is sufficient, system accepts the job, creates a job record, assigns a job ID, and forwards the job to the selected queue/device. 12. The user go to the printer and second verify at the printer using their ID card or user credentials (user name + password). 13. System returns confirmation to the user including job ID and estimated status. 14. The job is printed when the printer is available; job status updates are recorded.
Alternate Flows	**A1 — Insufficient quota:** After step 4, if the user's remaining quota is insufficient, the system rejects the job and informs the user (optionally offer to contact admin or request quota increase). **A2 — Invalid file / unsupported format:** If the document format is unsupported or corrupted, the system rejects the job and provides an error message. **A3 — Printer fault / unavailable:** If the target printer reports an error (paper out, low ink, offline), system notifies the user and suggests alternative queues (if available). The job may be held in a pending state. **A4 — Authentication failure:** If ID authentication fails, the job is not accepted until authentication succeeds. **A5 — User cancels:** User cancels before job submission; no job record is created. **A6 — Submitted file delayed:** The system should delete the submitted file and it should not stay in the queue for more than 24 hours.
Notes	System shall record timestamp, user ID, job ID, pages, cost estimate, and target device in the job record. System shall provide feedback to the user.

	Jobs should be traceable for auditing (who submitted, when, cost, and device info).

UC-03: View Own Printing History Log

Id	UC-03
Title	View Own Printing History Log
Primary Actor	Standard User
Goal	The user should be able to view their own printing jobs history log on the system dashboard.
Preconditions	- User is authenticated via Windows Active Directory. - User's account is not suspended.
Postconditions	- The user is presented with a filtered list of their print jobs and associated details (date/time, pages, title) -No system state is changed (read-only).
Basic Flow	1. User selects "Printing History" from the application or web interface. 2. System retrieves the user's print-job records from the database, optionally applying default filters (most recent first, last 30 days). 3. System displays the list with key details (job ID, date/time, pages, status) and provides controls for filtering, sorting, and exporting. 4. User reviews entries; user may select an entry to view full details (including file name, options, and logs).
Alternate Flows	**A1 — No history / empty result:** System displays a "no records" message and guidance (e.g., "You have no print jobs in this period"). **A2 — Authentication failure:** If Authentication session expired, user is prompted to re-authenticate.

	A3 — Partial data retrieval error: If the system fails to retrieve historical records, an error is shown and the user is advised to retry or contact Tech Support.
Notes	System shall ensure users can only view their own records (data isolation). Entries shall include job metadata required for auditing: job ID, timestamp, device, pages, cost, and final status.

UC-04: Add Printing Device

Id	UC-04
Title	Add Printing Device
Primary Actor	Admin
Goal	Add a new printing device to the system so it becomes available printing operations.
Preconditions	- Admin is authenticated via Windows Active Directory.
Postconditions	- A new printing device is added to the system configuration. - The device appears in the device list and can be used for print queue setup.
Basic Flow	1. Admin accesses the device management interface. 2. Admin selects **Add Printing Device**. 3. Admin enters the required device information (e.g., device name, IP address, location). 4. System validates the provided information. 5. System saves the device configuration and updates the device list. 6. System confirms successful operation to the Admin

Alternate Flows	**A1 — Invalid device configuration:** System displays validation errors and requests correction. **A2 — Duplicate device detected:** If a device with the same identifier (e.g., IP address/device name) already exists, the system rejects the request and notifies the Admin.
Notes	All device management actions shall be logged for auditing purposes.

UC-05: Remove Printing Device

Id	UC-05
Title	Remove Printing Device
Primary Actor	Admin
Goal	Remove an existing printing device from the system so it is no longer available for printing operations.
Preconditions	- Admin is authenticated via Windows Active Directory. - The target printing device exists in the system.
Postconditions	- The printing device is removed from the system configuration. - The device no longer appears in the device list or available queue/device selections.
Basic Flow	1. Admin accesses the device management interface. 2. Admin selects an existing printing device from the device list. 3. Admin selects **Remove Printing Device**. 4. System checks whether the device has active or pending print jobs.

	5. If allowed, the system removes the device configuration and updates the device list. 6. System confirms successful operation to the Admin.
Alternate Flows	**A1 — Device currently in use:** If the device has active or pending jobs, the system prevents removal and notifies the Admin. **A2 — Device not found:** If the selected device no longer exists, the system displays an error message.
Notes	All device management actions shall be logged for auditing purposes.

UC-06: Manage Printing Queues

Id	UC-06
Title	Manage Printing Queues
Primary Actor	Admin
Goal	Create Configure, or delete print queues
Preconditions	-Admin is authenticated. -At least one device exists in the system (for queue creation).
Postconditions	- Print queue configuration is updated accordingly.
Basic Flow	1. Admin accesses the queue management section. 2. Admin selects to create or delete a queue. 3. System validates request. 4. System updates queue configuration. 5. System confirms completion.

Alternate Flows	**A1 — Invalid queue configuration: System requests corrected input**
Notes	

UC-07: Modify User Quota

Id	UC-07
Title	Modify User Quota
Primary Actor	Admin, Technician
Goal	Update printing quota limits for users.
Preconditions	- Admin is authenticated. - Technician is authenticated - Target user account exists.
Postconditions	- Updated quota is applied immediately to the user account.
Basic Flow	1. Admin/Technician selects a user account. 2. Admin/Technician modifies printing quota value. 3. System validates quota value. 4. System updates user record. 5. System confirms update.
Alternate Flows	**A1 — Invalid quota value:** System rejects and requests valid input. **A2 — User not found:** System displays error message..
Notes	

UC-08: Manage User Access

Id	UC-08
Title	Manage User Access
Primary Actor	Admin, Technician
Goal	Suspend or reactivate user access to the system.
Preconditions	- Admin/Technician is authenticated. - Target user account exists.
Postconditions	- User account status is updated (active/suspended). - Suspended users cannot submit print jobs.
Basic Flow	1. Admin/Technician selects a user account. 2. Admin/Technician chooses to suspend or reactivate the user. 3. System updates user status. 4. System confirms action.
Alternate Flows	**A1 — Attempt to suspend another Admin: System denies action if restricted by policy.** **A2 — User not found: System displays error.**
Notes	

UC-09: View Logs, Cost Calculations, and Analytics Dashboard

Id	UC-09

Title	View Logs, Cost Calculations, and Analytics Dashboard
Primary Actor	Admin
Goal	Access system analytics, printing logs, and cost calculation reports.
Preconditions	- Admin is authenticated.
Postconditions	- Admin views up-to-date analytical and reporting data.
Basic Flow	1. Admin accesses the analytics/dashboard section. 2. System retrieves aggregated printing statistics and cost data. 3. System displays usage trends, user activity logs, device logs, and calculated costs. 4. Admin may apply filters (by date, department, device, or user). 5. System updates dashboard view accordingly.
Alternate Flows	**A1 — No data available: System displays appropriate message.** **A2 — Data retrieval error: System displays error and logs issue.**
Notes	Dashboard data shall be generated from system records. Cost calculations shall be accurate based on predefined pricing rules.

UC-10: Receive and View System Error Notification

Id	UC-10
Title	Issue System Notifications
Actors	Technician (Primary) / Printers (Secondary)
Goal	Allow Technical Support staff to view and review system or printer-related error notifications in order to take corrective action.
Preconditions	- Technical Support user is authenticated via Windows Active

	Directory. - System monitoring is active. - An error event has been detected and recorded by the system.
Postconditions	- Technical Support user has reviewed the error notification. - The error remains recorded in the system for tracking and auditing.
Basic Flow	1. Printer or system detects an operational issue (e.g., printer fault, no paper, low ink, system failure). 2. System records the error event. 3. System generates an error notification. 4. Technical Support accesses the notification interface. 5. System displays detailed error information (device identifier, timestamp, error type, status). 6. Technical Support reviews the notification and takes appropriate corrective action (within or outside the system).
Alternate Flows	**A1 — Notification delivery failure:** **If the notification cannot be delivered in real time, the system stores the notification for later viewing and logs the failure.** **A2 — Duplicate error events:** **If repeated identical errors occur within a short time interval, the system consolidates them into a single notification entry.**
Notes	Error notifications shall include device identifier, timestamp, and detailed error description. All error events shall be logged and retained for auditing and reporting purposes. Administrators may also have permission to view error logs, but resolution responsibility lies primarily with Technical Support. description. Notifications shall be stored for future reference.

UC-11: Modify Technician Privileges

Id	UC-11
Title	Modify Technician Privileges
Actors	Administrator (Primary)
Goal	Modify the system access privileges assigned to a Technical Support user.
Preconditions	- Administrator is authenticated via Windows Active Directory. - Target user account exists. - Target user has the Technical Support role.
Postconditions	- The technician's privileges are updated according to the Administrator's changes. - Updated permissions take effect immediately.
Basic Flow	1. Administrator accesses the user management interface. 2. Administrator selects a Technical Support user account. 3. Administrator views the current privilege settings. 4. Administrator modifies one or more privilege settings. 5. System validates the changes. 6. System updates the technician's privilege configuration. 7. System confirms successful update to the Administrator.
Alternate Flows	**A1 — Unauthorized Role Modification:** **If the Administrator attempts to modify privileges of another**

	Administrator, the system denies the request. **A2 — Invalid Privilege Configuration:** **If the selected privilege configuration is invalid or conflicts with system policy, the system rejects the update and requests correction.**
Notes	Only Administrators may modify technician privileges. All privilege modification actions shall be logged for auditing purposes.

Functional Requirements Specification

1. Authentication & Authorization

- **FR-1.1** The system shall authenticate users using Microsoft Windows Active Directory (AD).
- **FR-1.2** The system shall retrieve user identity attributes from AD (e.g., username, department, group membership) upon successful authentication.
- **FR-1.3** The system shall authorize users based on their assigned role (Administrator, Technical Support, Standard User).
- **FR-1.4** The system shall restrict access to system features according to role permissions.
- **FR-1.5** The system shall prevent suspended users from accessing the system even if AD authentication succeeds.
- **FR-1.6** The system shall log all authentication attempts for auditing purposes.

2. Print Job Management (Standard User)

- **FR-2.1** The system shall allow Standard Users to submit print jobs to an available print queue.
- **FR-2.2** The system shall validate the user's available printing quota before accepting a print job.
- **FR-2.3** The system shall reject print jobs if the user's quota is insufficient.
- **FR-2.4** The system shall generate a unique job identifier for each submitted print job.
- **FR-2.5** The system shall forward accepted print jobs to the selected printing device.
- **FR-2.6** The system shall provide confirmation or error feedback after job submission.
- **FR-2.7** The system shall record job details including user ID, timestamp, pages, device, status, and calculated cost.
- **FR-2.8** The system shall allow users to cancel a pending print job.

3. Print History

- **FR-3.1** The system shall allow Standard Users to view their own printing history log.
- **FR-3.2** The system shall prevent users from accessing other users' printing history records.
- **FR-3.3** The system shall display job metadata including job ID, date/time, device, number of pages, status, and calculated cost.

4. Device & Queue Management (Administrator)

- FR-4.1 The system shall allow Administrators to add new printing devices.
- FR-4.2 The system shall allow Administrators to remove existing printing devices.
- FR-4.3 The system shall allow Administrators to create and configure print queues.
- FR-4.4 The system shall allow Administrators to add new printing devices.
- FR-4.5 The system shall allow Administrators to delete print queues
- FR-4.6 The system shall allow deletion of devices or queues that contain active print jobs.

5. User & Quota Management (Administrator & Technician)

- FR-5.1 The system shall allow Administrators to modify user printing quotas.
- FR-5.2 The system shall allow Technical Support staff to modify user printing quotas.
- FR-5.3 The system shall allow Administrators to suspend or reactivate user accounts.
- FR-5.4 The system shall allow Technical Support staff to suspend or reactivate user accounts.
- FR-5.5 The system shall prevent Technical Support staff from modifying Administrator accounts.
- FR-5.6 The system shall log all quota and access modifications for auditing purposes.

6. User & Quota Management (Administrator & Technician)

- FR-6.1 The system shall record detailed printing activity logs for users and devices.
- FR-6.2 The system shall allow Administrators to access user and device printing history logs.
- FR-6.3 The system shall detect printer-related errors including low paper, low ink, device faults, and offline status.
- FR-6.4 The system shall generate system error notifications when operational failures occur.
- FR-6.5 The system shall notify Technical Support staff of detected system or printer errors.
- FR-6.6 The system shall store error notifications for future review.

7. User & Quota Management (Administrator & Technician)

- FR-7.1 The system shall calculate the cost of each print job based on predefined pricing rules.
- FR-7.2 The system shall allow Administrators to access user and device printing history logs.
- FR-7.3 The system shall detect printer-related errors including low paper, low ink, device faults, and offline status.
- FR-7.4 The system shall generate system error notifications when operational failures occur.

8. Role & Privilege Management

- **FR-8.1** The system shall allow Administrators to modify the system privileges assigned to Technical Support users.

- **FR-8.2** The system shall prevent Technical Support users from modifying their own privileges.
- **FR-8.3** The system shall prevent Administrators from modifying privileges of other Administrators (unless explicitly allowed by system policy).
- **FR-8.4** The system shall log all privilege modification actions for auditing purposes

Non-Functional Requirements Specification

1. Performance

- NFR-1.1 The system shall process and validate standard print job submissions within 3 seconds when the client-to-server network latency is ≤ 150 ms.
- NFR-1.2 The system shall automatically purge stuck or unreleased print files from the server after 24 hours to optimize storage and memory.
- NFR-1.3 The system shall support at least 200 concurrent active users (faculty, staff, and students) during peak campus hours while keeping the dashboard initial load time ≤ 3 seconds
- NFR-1.4 The system shall achieve ≥ 99.5% uptime during working hours (Sunday-Thursday, 8:00 AM-5:00 PM), excluding scheduled maintenance.

2. Usability

- NFR-2.1 The system's web interface shall allow Standard Users to navigate to their printing history and view logs within two clicks from the main dashboard.
- NFR-2.2 The system shall visually separate print queues available to staff from those available to students to prevent user confusion and submission errors.

3. Security

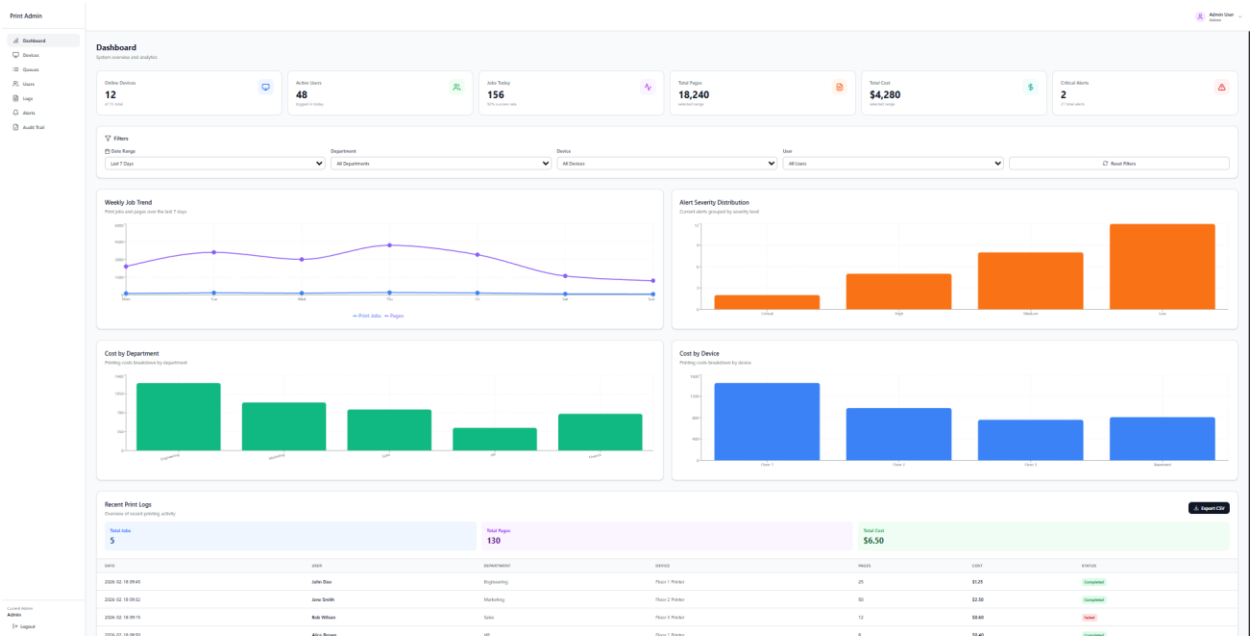
- NFR-3.1 The system shall securely synchronize user details, groups, and departments from Active Directory at least once every 24 hours to ensure access control lists remain perfectly up to date.
- NFR-3.2 All network communications between the user clients, the Print Management System, and the hardware printers shall be encrypted using TLS 1.2 or higher.

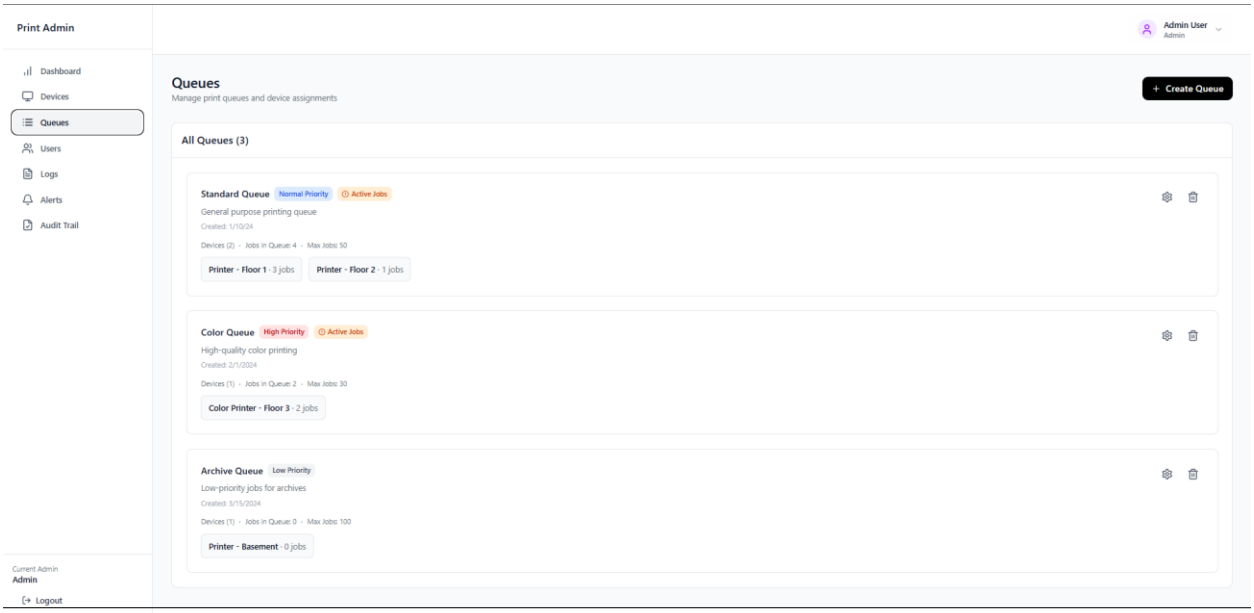
4. Compatibility & Constraints

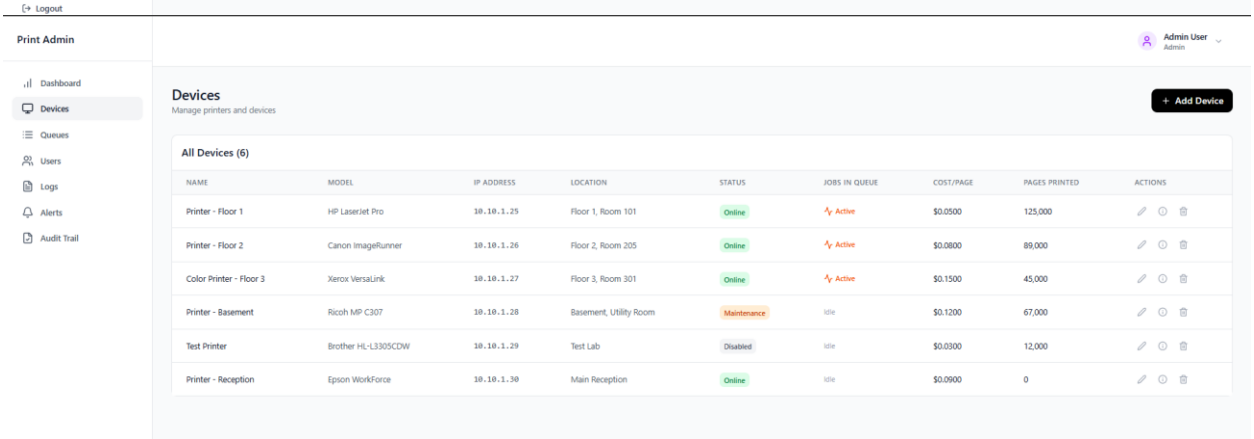
- NFR-4.1 The system shall natively integrate with the HP OXP (Open Extensibility Platform) hardware commonly used in the college.
- NFR-4.2 The system shall support deployment and printer management exclusively through the existing Windows Server environment.
- NFR-4.3 The system's user and admin web interfaces shall be fully responsive and functional across the latest two major versions of Chrome, Edge, and Safari on both desktop and mobile devices.

Wireframe (Figma)

1. **Admin Dashboard**
   [https://www.figma.com/make/MmscdtPARrAD3rqVo5Pp8A/Admin-PMS?t=Y020Pn1gOPumyxp8-1]




Print Admin
Admin User
Admin
Dashboard
Devices
Queues
Users
Logs
Alerts
Audit Trail
Queues
Manage print queues and device assignments
+ Create Queue
All Queues (3)
Standard Queue
Normal Priority
Active Jobs
General purpose printing queue
Created: 1/10/24
Devices (2) · Jobs in Queue: 4 · Max Jobs: 50
Printer - Floor 1 · 3 jobs
Printer - Floor 2 · 1 jobs
Color Queue
High Priority
Active Jobs
High-quality color printing
Created: 2/1/2024
Devices (1) · Jobs in Queue: 2 · Max Jobs: 30
Color Printer - Floor 3 · 2 jobs
Archive Queue
Low Priority
Low-priority jobs for archives
Created: 3/15/2024
Devices (1) · Jobs in Queue: 0 · Max Jobs: 100
Printer - Basement · 0 jobs
Current Admin
Admin
Logout


Print Admin
Admin User
Admin
Dashboard
Devices
Queues
Users
Logs
Alerts
Audit Trail
Devices
Manage printers and devices
+ Add Device
All Devices (6)
NAME
MODEL
IP ADDRESS
LOCATION
STATUS
JOBS IN QUEUE
COST/PAGE
PAGES PRINTED
ACTIONS
Printer - Floor 1
HP LaserJet Pro
10.10.1.25
Floor 1, Room 101
Online
Active
$0.0500
125,000
Printer - Floor 2
Canon ImageRunner
10.10.1.26
Floor 2, Room 205
Online
Active
$0.0800
89,000
Color Printer - Floor 3
Xerox VersaLink
10.10.1.27
Floor 3, Room 301
Online
Active
$0.1500
45,000
Printer - Basement
Ricoh MP C307
10.10.1.28
Basement, Utility Room
Maintenance
Idle
$0.1200
67,000
Test Printer
Brother HL-L3305CDW
10.10.1.29
Test Lab
Disabled
Idle
$0.0300
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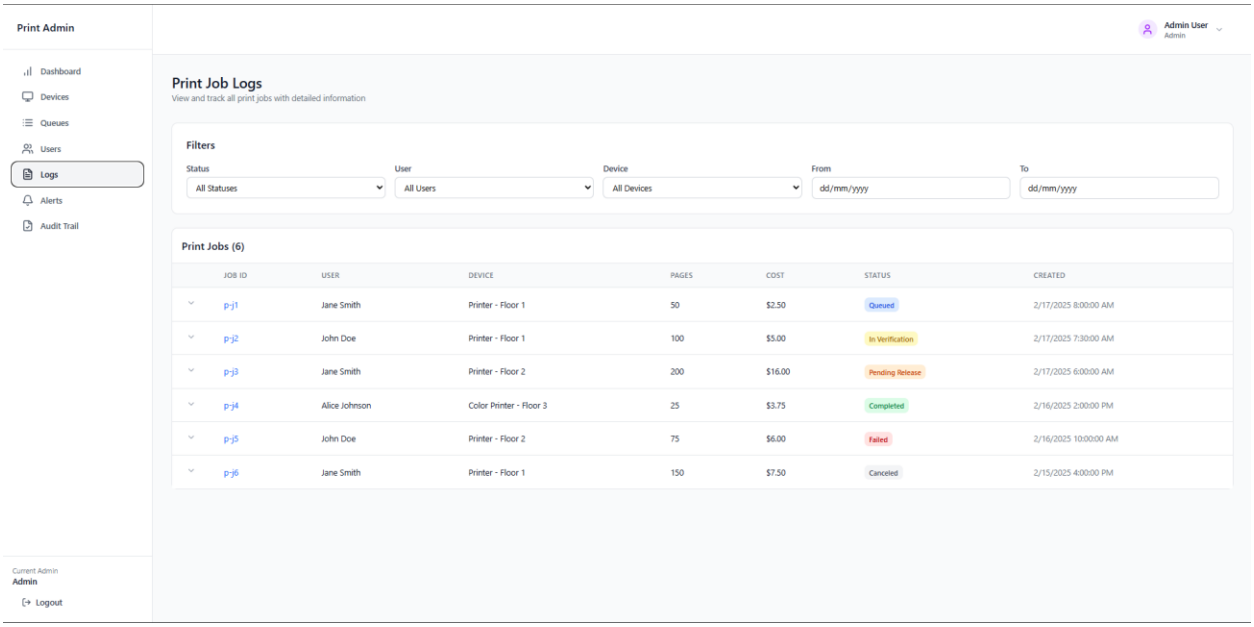
Printer - Reception
Epson WorkForce
10.10.1.30
Main Reception
Online
Idle
$0.0900
0


Print Admin
Admin User
Admin
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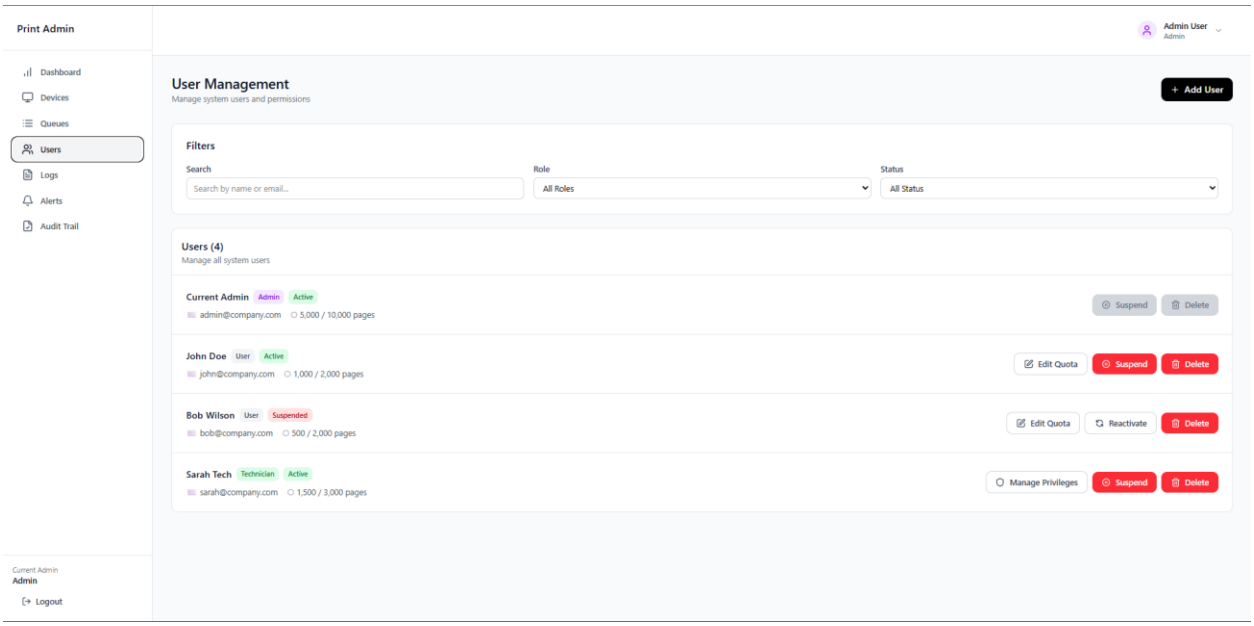
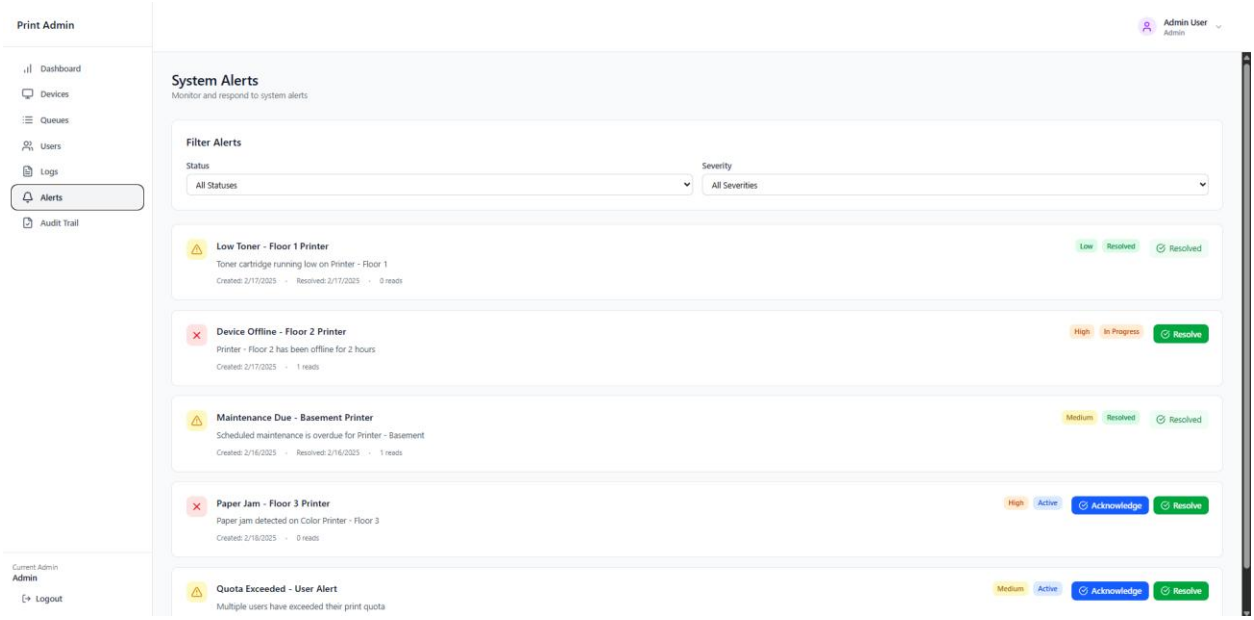
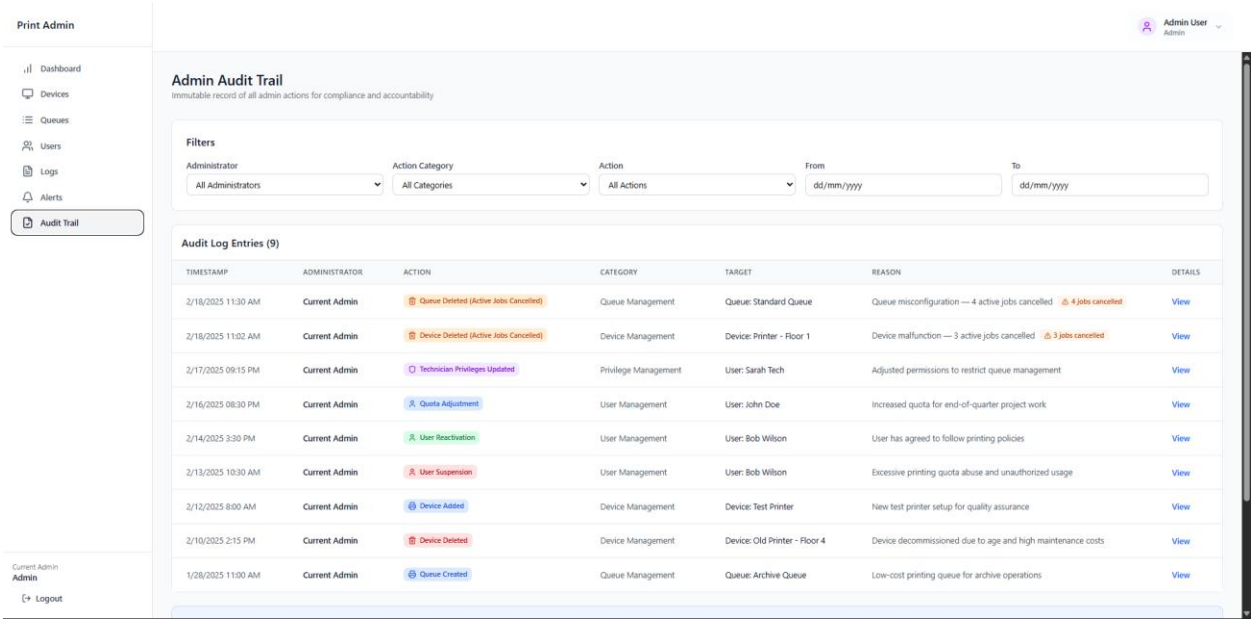
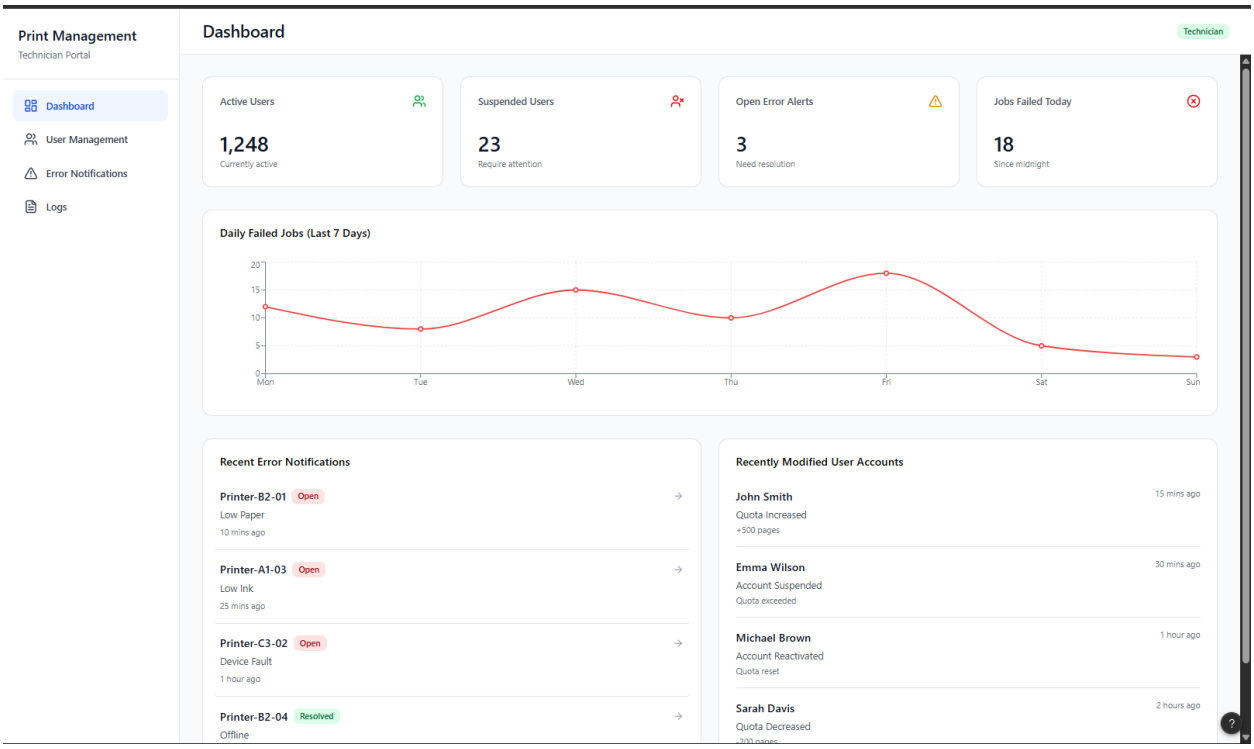
Devices
Queues
Users
Logs
Alerts
Audit Trail
Print Job Logs
View and track all print jobs with detailed information
Filters
Status
All Statuses
User
All Users
Device
All Devices
From
dd/mm/yyyy
To
dd/mm/yyyy
Print Jobs (6)
JOB ID
USER
DEVICE
PAGES
COST
STATUS
CREATED
p-j1 Jane Smith Printer - Floor 1 50 $2.50 Queued 2/17/2025 8:00:00 AM
p-j2 John Doe Printer - Floor 1 100 $5.00 In Verification 2/17/2025 7:30:00 AM
p-j3 Jane Smith Printer - Floor 2 200 $16.00 Pending Release 2/17/2025 6:00:00 AM
p-j4 Alice Johnson Color Printer - Floor 3 25 $3.75 Completed 2/16/2025 2:00:00 PM
p-j5 John Doe Printer - Floor 2 75 $6.00 Failed 2/16/2025 10:00:00 AM
p-j6 Jane Smith Printer - Floor 1 150 $7.50 Canceled 2/15/2025 4:00:00 PM
Current Admin
Admin
Logout

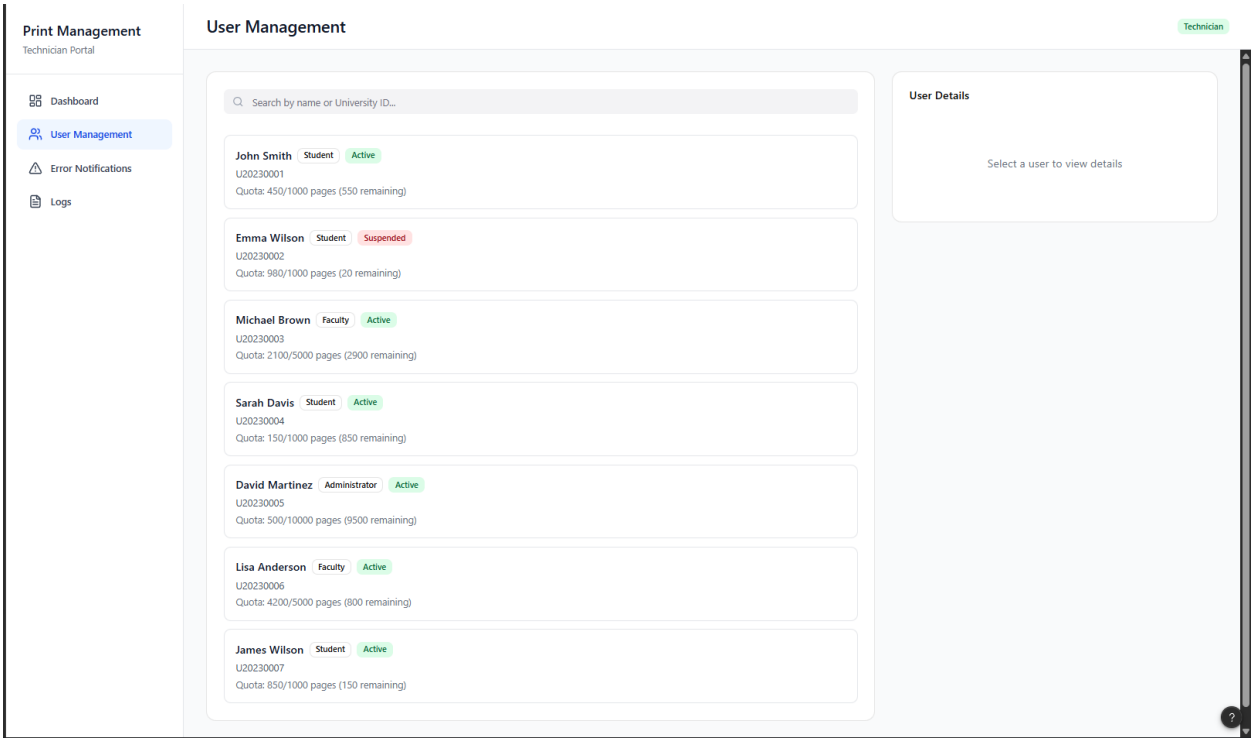

Print Admin
Admin User
Admin
Dashboard
Devices
Queues
Users
Logs
Alerts
Audit Trail
User Management
Manage system users and permissions
+ Add User
Filters
Search
Search by name or email...
Role
All Roles
Status
All Status
Users (4)
Manage all system users
Current Admin Admin Active
admin@company.com 5,000 / 10,000 pages
Suspend Delete
John Doe User Active
john@company.com 1,000 / 2,000 pages
Edit Quota Suspend Delete
Bob Wilson User Suspended
bob@company.com 500 / 2,000 pages
Edit Quota Reactivate Delete
Sarah Tech Technician Active
sarah@company.com 1,500 / 3,000 pages
Manage Privileges Suspend Delete
Current Admin
Admin
Logout

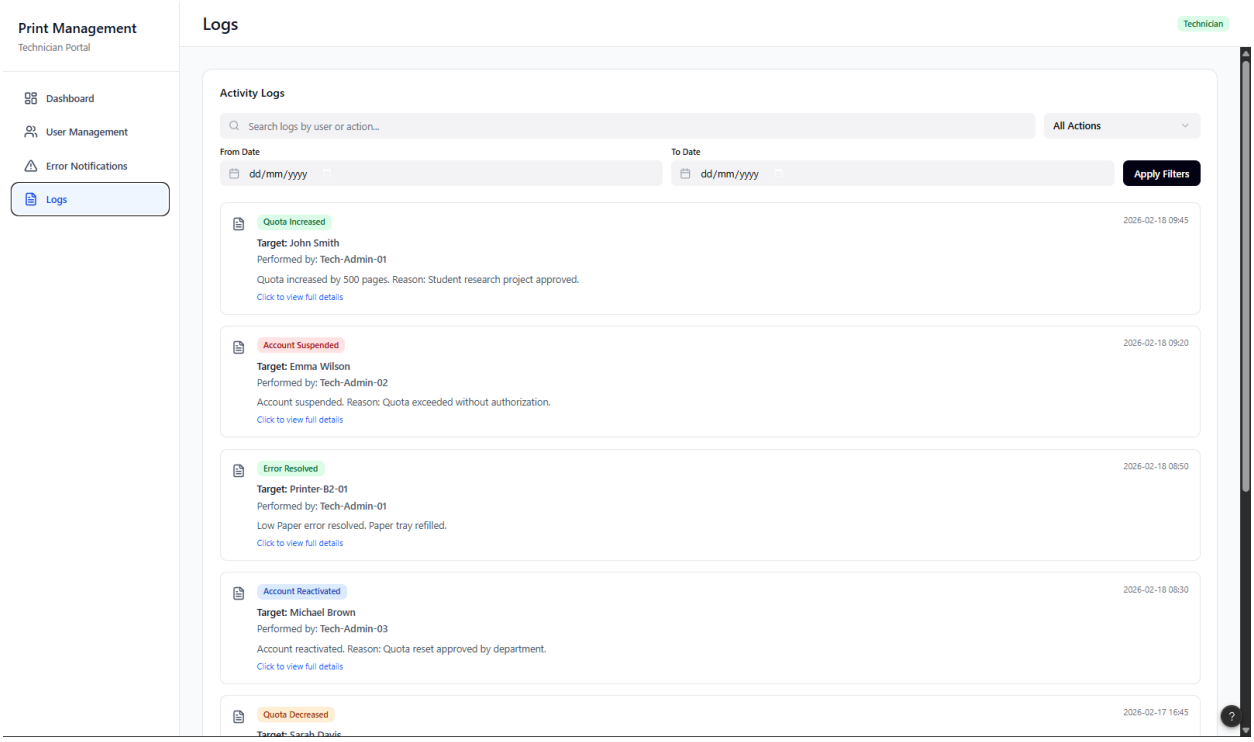

Print Admin
Admin User
Admin
Dashboard
Devices
Queues
Users
Logs
Alerts
Audit Trail
System Alerts
Monitor and respond to system alerts
Filter Alerts
Status
All Statuses
Severity
All Severities
Low Toner - Floor 1 Printer
Toner cartridge running low on Printer - Floor 1
Created: 2/17/2025 · Resolved: 2/17/2025 · 0 reads
Low
Resolved
Resolved
Device Offline - Floor 2 Printer
Printer - Floor 2 has been offline for 2 hours
Created: 2/17/2025 · 1 reads
High
In Progress
Resolve
Maintenance Due - Basement Printer
Scheduled maintenance is overdue for Printer - Basement
Created: 2/16/2025 · Resolved: 2/16/2025 · 1 reads
Medium
Resolved
Resolved
Paper Jam - Floor 3 Printer
Paper jam detected on Color Printer - Floor 3
Created: 2/18/2025 · 0 reads
High
Active
Acknowledge
Resolve
Quota Exceeded - User Alert
Multiple users have exceeded their print quota
Medium
Active
Acknowledge
Resolve
Current Admin
Admin
Logout


Print Admin
Admin User
Admin
Dashboard
Devices
Queues
Users
Logs
Alerts
Audit Trail
Admin Audit Trail
Immutable record of all admin actions for compliance and accountability
Filters
Administrator
All Administrators
Action Category
All Categories
Action
All Actions
From
dd/mm/yyyy
To
dd/mm/yyyy
Audit Log Entries (9)
TIMESTAMP
ADMINISTRATOR
ACTION
CATEGORY
TARGET
REASON
DETAILS
2/18/2025 11:30 AM
Current Admin
Queue Deleted (Active Jobs Cancelled)
Queue Management
Queue: Standard Queue
Queue misconfiguration — 4 active jobs cancelled
4 jobs cancelled
View
2/18/2025 11:02 AM
Current Admin
Device Deleted (Active Jobs Cancelled)
Device Management
Device: Printer - Floor 1
Device malfunction — 3 active jobs cancelled
3 jobs cancelled
View
2/17/2025 09:15 PM
Current Admin
Technician Privileges Updated
Privilege Management
User: Sarah Tech
Adjusted permissions to restrict queue management
View
2/16/2025 08:30 PM
Current Admin
Quota Adjustment
User Management
User: John Doe
Increased quota for end-of-quarter project work
View
2/14/2025 3:30 PM
Current Admin
User Reactivation
User Management
User: Bob Wilson
User has agreed to follow printing policies
View
2/13/2025 10:30 AM
Current Admin
User Suspension
User Management
User: Bob Wilson
Excessive printing quota abuse and unauthorized usage
View
2/12/2025 8:00 AM
Current Admin
Device Added
Device Management
Device: Test Printer
New test printer setup for quality assurance
View
2/10/2025 2:15 PM
Current Admin
Device Deleted
Device Management
Device: Old Printer - Floor 4
Device decommissioned due to age and high maintenance costs
View
1/28/2025 11:00 AM
Current Admin
Queue Created
Queue Management
Queue: Archive Queue
Low-cost printing queue for archive operations
View
Current Admin
Admin
Logout

2. Technician Dashboard [https://www.figma.com/make/zXV5usocwXjqTyjLRz0ent/Technician-PMS?t=AhgwHPxhzLe2m3Az-1]

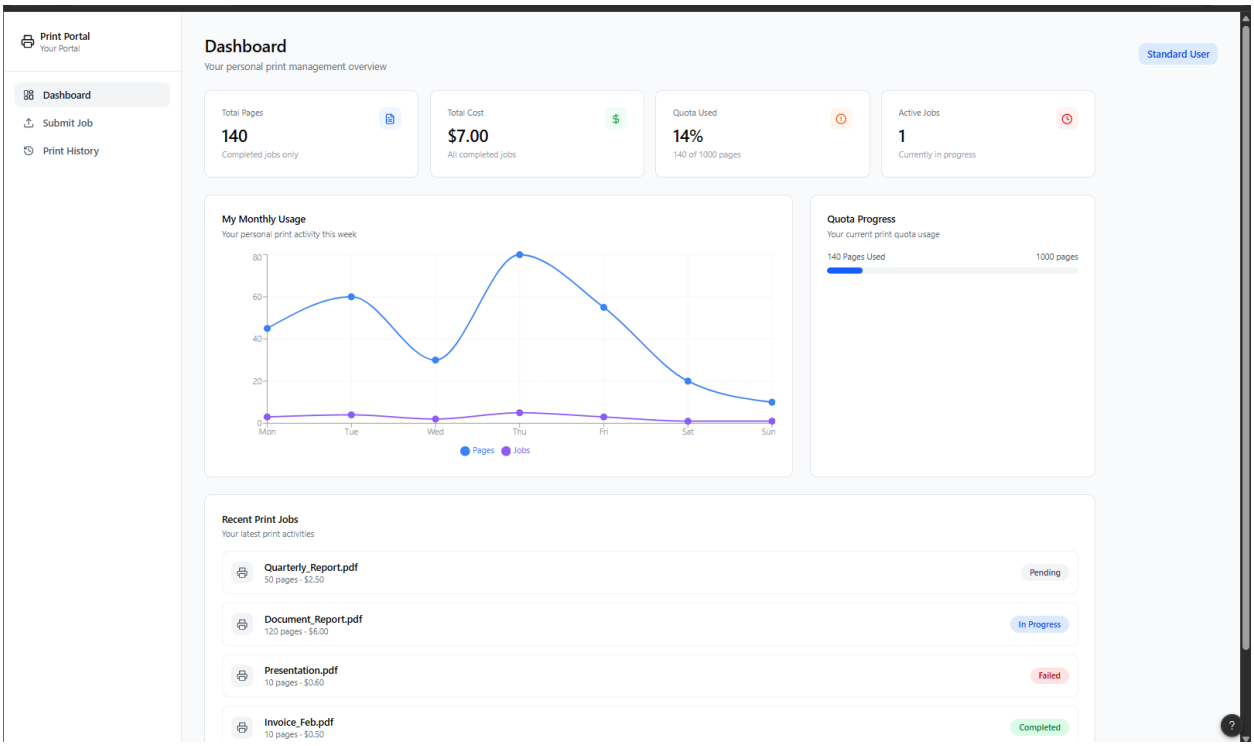


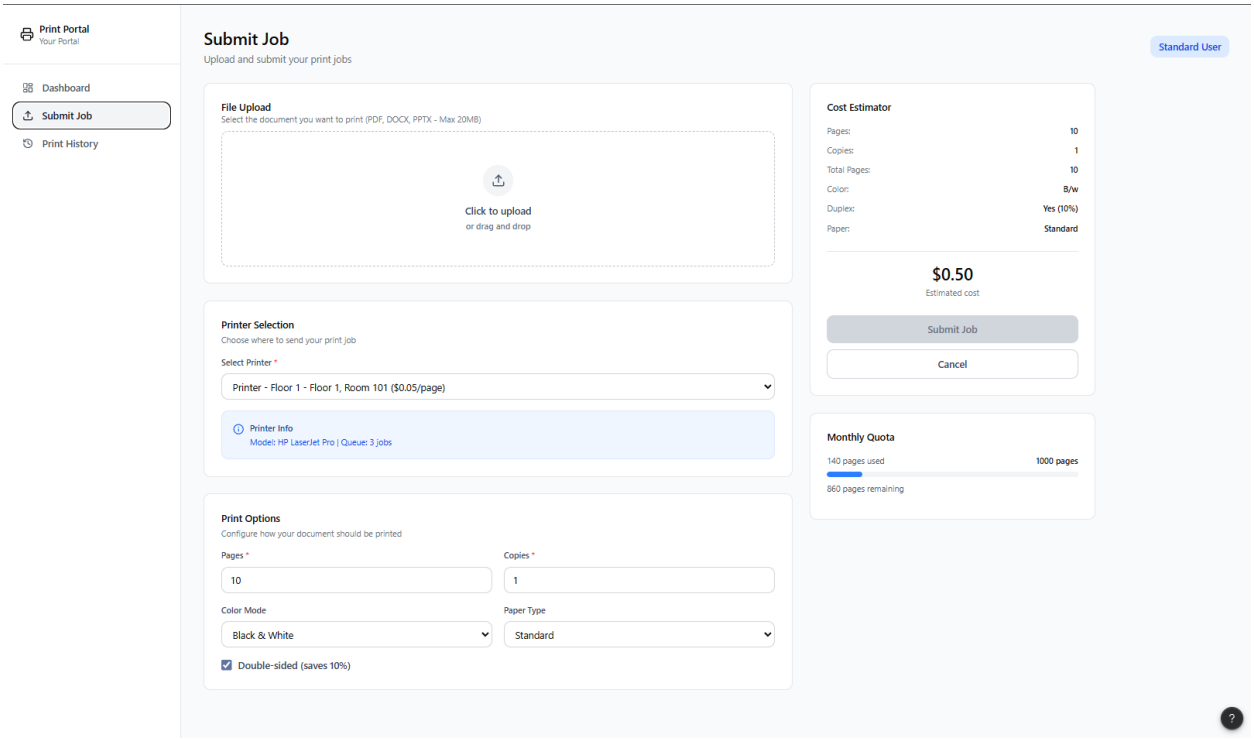

Print Management
Technician Portal
Dashboard
User Management
Error Notifications
Logs
User Management
Technician
Search by name or University ID...
John Smith Student Active
U20230001
Quota: 450/1000 pages (550 remaining)
Emma Wilson Student Suspended
U20230002
Quota: 980/1000 pages (20 remaining)
Michael Brown Faculty Active
U20230003
Quota: 2100/5000 pages (2900 remaining)
Sarah Davis Student Active
U20230004
Quota: 150/1000 pages (850 remaining)
David Martinez Administrator Active
U20230005
Quota: 500/10000 pages (9500 remaining)
Lisa Anderson Faculty Active
U20230006
Quota: 4200/5000 pages (800 remaining)
James Wilson Student Active
U20230007
Quota: 850/1000 pages (150 remaining)
User Details
Select a user to view details


Print Management
Technician Portal
Dashboard
User Management
Error Notifications
Logs
Logs
Technician
Activity Logs
Search logs by user or action...
All Actions
From Date
dd/mm/yyyy
To Date
dd/mm/yyyy
Apply Filters
Quota Increased
2026-02-18 09:45
Target: John Smith
Performed by: Tech-Admin-01
Quota increased by 500 pages. Reason: Student research project approved.
Click to view full details
Account Suspended
2026-02-18 09:20
Target: Emma Wilson
Performed by: Tech-Admin-02
Account suspended. Reason: Quota exceeded without authorization.
Click to view full details
Error Resolved
2026-02-18 08:50
Target: Printer-B2-01
Performed by: Tech-Admin-01
Low Paper error resolved. Paper tray refilled.
Click to view full details
Account Reactivated
2026-02-18 08:30
Target: Michael Brown
Performed by: Tech-Admin-03
Account reactivated. Reason: Quota reset approved by department.
Click to view full details
Quota Decreased
2026-02-17 16:45

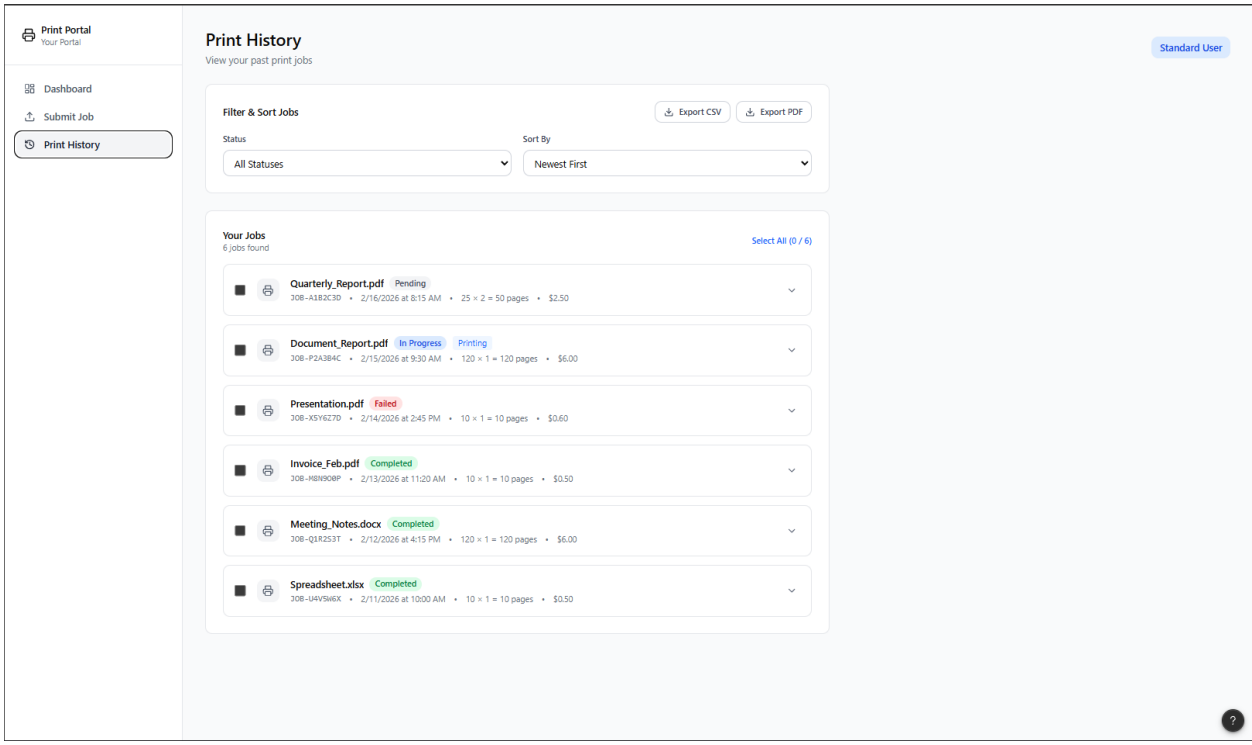
3. **Users Dashboard**

[https://www.figma.com/make/dOfrNMeNw1vybturf2cuG4/User-PMS?t=AhgwHPxhzLe2m3Az-1]

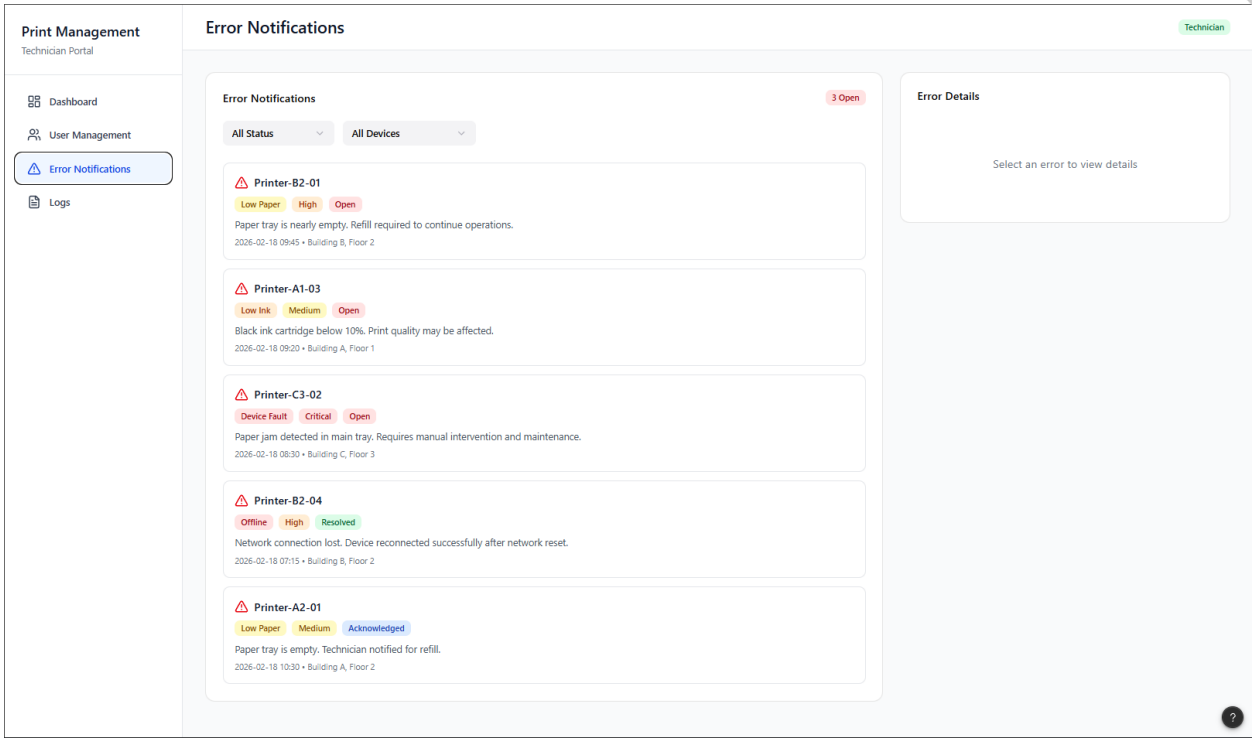



Print Portal
Your Portal
Dashboard
Submit Job
Print History
Submit Job
Upload and submit your print jobs
Standard User
File Upload
Select the document you want to print (PDF, DOCX, PPTX - Max 20MB)
Click to upload
or drag and drop
Printer Selection
Choose where to send your print job
Select Printer *
Printer - Floor 1 - Floor 1, Room 101 ($0.05/page)
Printer Info
Model: HP LaserJet Pro | Queue: 3 jobs
Print Options
Configure how your document should be printed
Pages *
10
Copies *
1
Color Mode
Black & White
Paper Type
Standard
Double-sided (saves 10%)
Cost Estimator
Pages: 10
Copies: 1
Total Pages: 10
Color: B/w
Duplex: Yes (10%)
Paper: Standard
$0.50
Estimated cost
Submit Job
Cancel
Monthly Quota
140 pages used
1000 pages
860 pages remaining

1. Figma Project Link [https://w



ww.figma.com/files/team/1369054191536550891/project/230760201?fuid=136905418917656


Print Management
Technician Portal
Dashboard
User Management
Error Notifications
Logs
Error Notifications
Technician
Error Notifications
3 Open
All Status
All Devices
Printer-B2-01
Low Paper
High
Open
Paper tray is nearly empty. Refill required to continue operations.
2026-02-18 09:45 • Building B, Floor 2
Printer-A1-03
Low Ink
Medium
Open
Black ink cartridge below 10%. Print quality may be affected.
2026-02-18 09:20 • Building A, Floor 1
Printer-C3-02
Device Fault
Critical
Open
Paper jam detected in main tray. Requires manual intervention and maintenance.
2026-02-18 08:30 • Building C, Floor 3
Printer-B2-04
Offline
High
Resolved
Network connection lost. Device reconnected successfully after network reset.
2026-02-18 07:15 • Building B, Floor 2
Printer-A2-01
Low Paper
Medium
Acknowledged
Paper tray is empty. Technician notified for refill.
2026-02-18 10:30 • Building A, Floor 2
Error Details
Select an error to view details

3017]